

1	Foundations of Probability	
1	Probability basics and terminology	
2	Random experiments, sample spaces, and events	
3	Classical, empirical, and subjective probability	
4	Axioms of probability	
5	Complement, union, and intersection of events	
6	Mutually exclusive and exhaustive events	
7	Addition rule	
8	Multiplication rule	
9	Permutations and combinations	
10	Odds vs probability	

2	Conditional Probability and Bayes' Theorem	
1	Law of Large Numbers	
2	Central Limit Theorem	
3	Conditional probability	
4	Independent and dependent events	
5	Law of total probability	
6	Bayes' theorem	
7	Prior, likelihood, and posterior	
8	Tree diagrams	
9	Naive Bayes intuition	
10	Likelihood vs probability	

3	Random Variables	
1	Random variables (discrete and continuous)	
2	Probability mass function (PMF)	
3	Probability density function (PDF)	
4	Cumulative distribution function (CDF)	
5	Expected value and Variance	
6	Bernoulli and Binomial distributions	
7	Poisson and Geometric distributions	
8	Uniform and Exponential distributions	
9	Normal distribution	
10	Standard normal distribution and z-scores	